## 1 Label-free node representation learning

Input graph $\mathcal{V}, \mathcal{E}, \mathbf{X}$

Node features $\mathbf{X}$

Adjacency $\mathbf{A}$

| | $v_1$ | $v_2$ | $\cdots$ | $v_N$ |
|---|---|---|---|---|
| $v_1$ | 0 | 1 | $\cdots$ | 0 |
| $v_2$ | 1 | 0 | $\cdots$ | 1 |
| $\cdots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ |
| $v_N$ | 0 | 1 | $\cdots$ | 0 |

Two augmented views

View 1: edge dropping + feature masking

View 2: independently sampled

Frozen encoder on the original graph

Shared graph encoder (GRACE)

Projection head

View 1 projection

View 2 projection

Frozen node representation $\mathbf{G}$

## 2 Label-agnostic granular ball partition

Geometry-only partition in frozen representation space $\mathbf{G}$

Initialization $\mathcal{B} = \{\mathcal{V}\}$

Recursive 2-means split

Further splitting

Final granular ball partition in $\mathbf{G}$

selected ball $B$

Partition all nodes without using labels.

## 3 Leave-one-out local-consensus targets

(a) Selected granular ball $B$

v

v's own vote removed

(b) Leave-one-out consensus

$\mathbf{r}$ (excluding $v$)

$$\mathbf{q} = (1-\beta)\mathbf{r} + \beta\mathbf{e}_{i^\star}$$

If the highest vote count is tied, use $\mathbf{q} = \mathbf{r}$.

(c) Construct a soft target $\mathbf{t}$

Observed label $(1-\alpha)\mathbf{e}_z$

Local consensus $\alpha\mathbf{q}$

Soft target $\mathbf{t}$

$$\mathbf{t} = (1-\alpha)\mathbf{e}_z + \alpha\mathbf{q}$$

$\alpha = (C \max_i r_i - 1)/(C-1)$

Fewer than $s_{\min}$ remaining votes: retain the observed label.

## 4 MLP training and graph-free inference

MLP training with GB-LCSS targets

node feature $\mathbf{x}$

2-layer MLP $f$

softmax

prediction $\mathbf{p}$

Soft cross-entropy $-\sum_{i=1}^{C} t_i \log p_i$

Update MLP parameters

Graph-free inference (transductive)

test-node feature $\mathbf{x}$

Trained MLP $f$

softmax

class probabilities $\mathbf{p}$

Predicted class: 3 $\arg\max_i p_i = 3$

Observed noisy labels C1 C2 C3 C4 C5 non-voting node excluded self-vote data flow symmetric contrastive loss **Inference input**: $\mathbf{x}$ only